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Kelas : IF-1

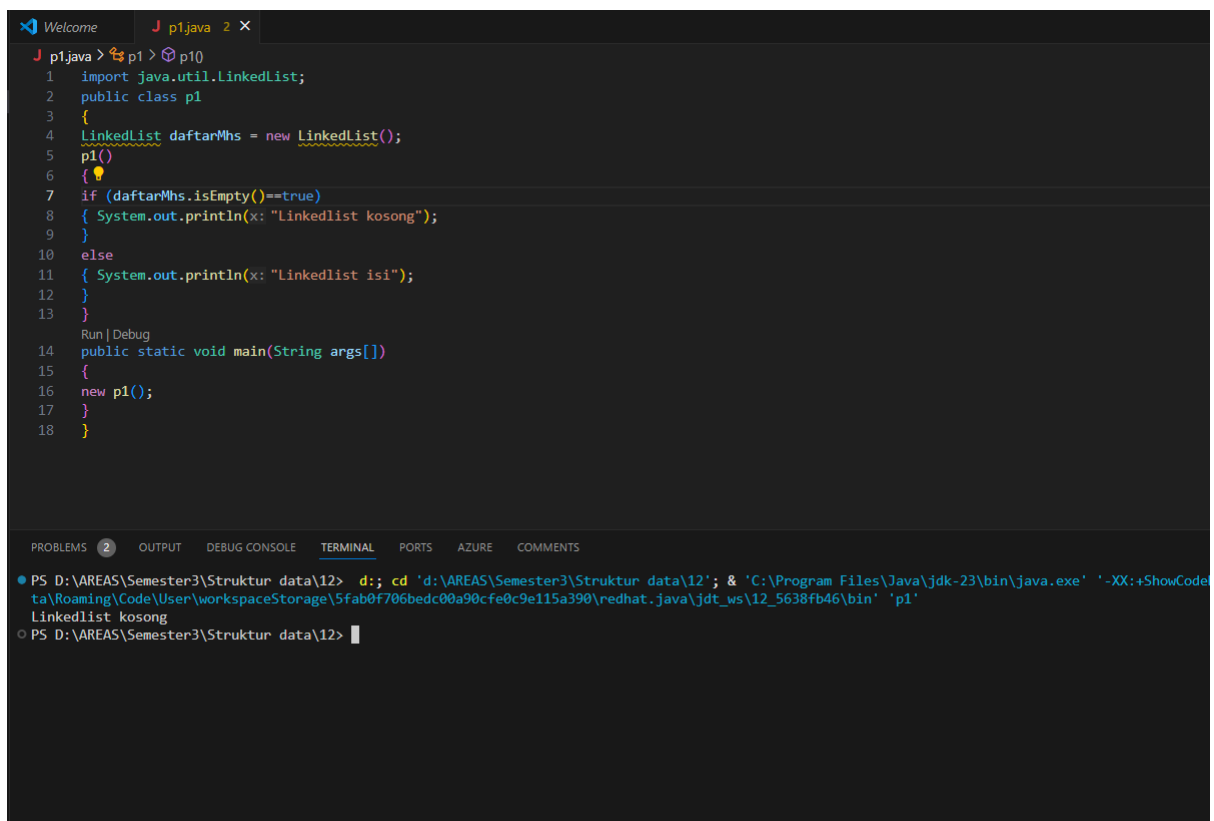
LISTING PRAKTIKUM STRUKTUR DATA

MODUL 12

COLLECTION

PRAKTIK

Praktik 1

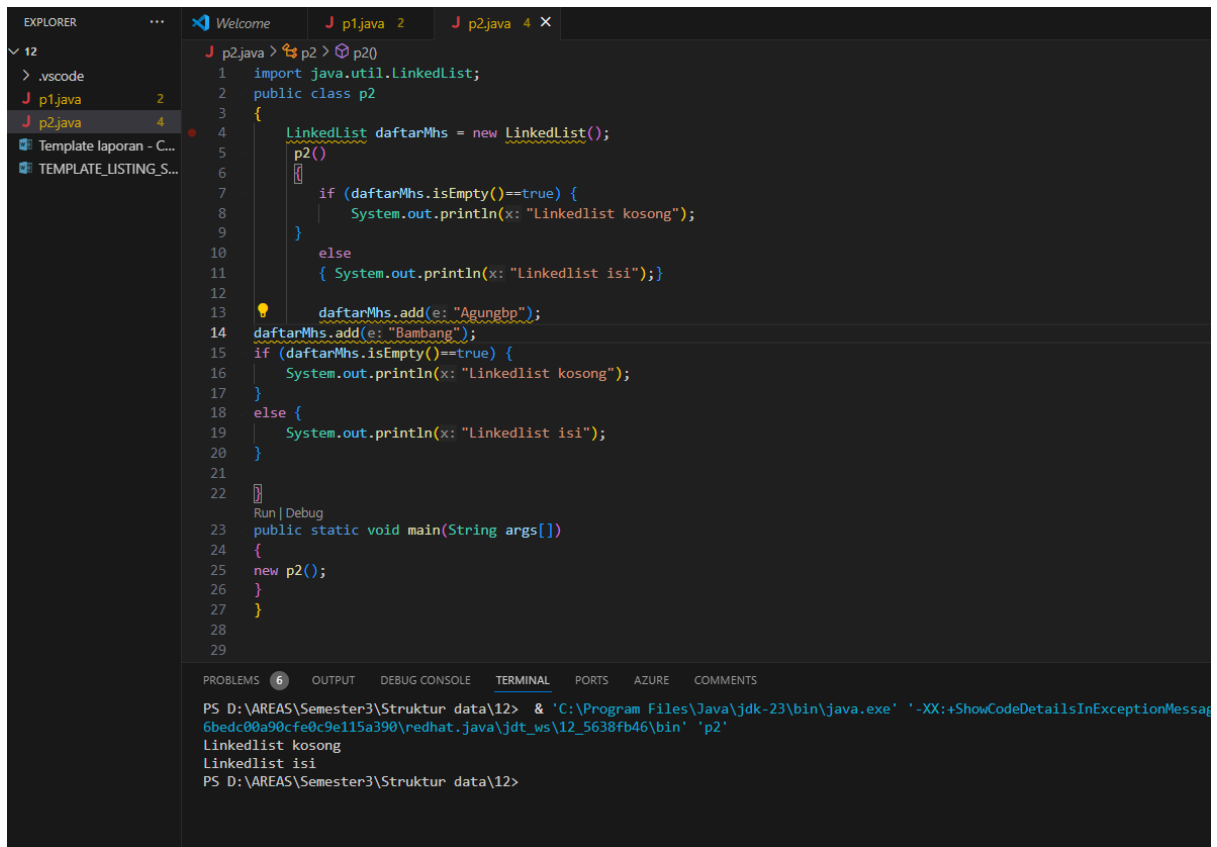


```

Welcome
J p1.java 2 X
J p1.java > p1 > p10
1  import java.util.LinkedList;
2  public class p1
3  {
4      LinkedList daftarMhs = new LinkedList();
5      p1()
6      {
7          if (daftarMhs.isEmpty()==true)
8          { System.out.println(x: "Linkedlist kosong");
9          }
10         else
11         { System.out.println(x: "Linkedlist isi");
12         }
13     }
14     Run | Debug
15     public static void main(String args[])
16     {
17         new p1();
18     }
19 }

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE COMMENTS
● PS D:\AREAS\Semester3\Struktur data\12> d:; cd 'd:\AREAS\Semester3\Struktur data\12'; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCode
ta\Roaming\Code\User\workspaceStorage\5fab0f706bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p1'
Linkedlist kosong
○ PS D:\AREAS\Semester3\Struktur data\12> 
```

Praktik 2



```
1 import java.util.LinkedList;
2 public class p2
3 {
4     LinkedList daftarMhs = new LinkedList();
5     p2()
6     {
7         if (daftarMhs.isEmpty()==true) {
8             System.out.println(x: "Linkedlist kosong");
9         }
10        else
11        { System.out.println(x: "Linkedlist isi");}
12    }
13    daftarMhs.add(e: "Agungbp");
14    daftarMhs.add(e: "Bambang");
15    if (daftarMhs.isEmpty()==true) {
16        System.out.println(x: "Linkedlist kosong");
17    }
18    else {
19        System.out.println(x: "Linkedlist isi");
20    }
21 }
22
23 public static void main(String args[])
24 {
25     new p2();
26 }
27 }
28
29
```

Run | Debug

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE COMMENTS

PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessage' 6bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p2'

Linkedlist kosong

Linkedlist isi

PS D:\AREAS\Semester3\Struktur data\12>

Praktik 3

```
import java.util.LinkedList ;
import java.util.tinkedLis ;
public class p3  t
{
    LinkedList  daftarMhs = new LinkedList ();
    p3()          t

    {
        daftarMhs.add("Agungbp");
        daftarMhs.add("Bambang");
        daftarMhs.add("Cucuk");
        daftarMhs.add("Dion");
        daftarMhs.add("Ending");
        daftarMhs.add("Fifin");
        daftarMhs.add("Gesit");
        daftarMhs.add("Heru");
        daftarMhs.add("Irma");
        daftarMhs.add("Janti");
        System.out.println(daftarMhs.get(0));
        System.out.println(daftarMhs.get(1));
        System.out.println(daftarMhs.get(2));
        System.out.println(daftarMhs.get(3));
        System.out.println(daftarMhs.get(4));
        System.out.println(daftarMhs.get(5));
        System.out.println(daftarMhs.get(6));
        System.out.println(daftarMhs.get(7));
        System.out.println(daftarMhs.get(8));
        System.out.println(daftarMhs.get(9));
        System.out.println("");
    }
    public static void main(String args[])
    {
        new p3();
    }
}
```

```
● PS D:\AREAS\Semester3\Struktur data\12> d:; cd 'd:\AREAS\Semester3\Struktur data\Roaming\Code\User\workspaceStorage\5fab0f706bedc00a90cfe0c9e115a390\n
Agungbp
Bambang
Cucuk
Dion
Ending
Fifin
Gesit
Heru
Irma
Janti

○ PS D:\AREAS\Semester3\Struktur data\12> █
```

Praktik 4

```
Welcome | Snapcode | p4.java 9+ X
J p4.java > p4 > p4()
1  import java.util.LinkedList;
2  public class p4
3  {
4      LinkedList daftarMhs = new LinkedList();
5  p4()
6  {
7      daftarMhs.add(e: "Agungbp");
8      daftarMhs.add(e: "Bambang");
9      daftarMhs.add(e: "Cucuk");
10     daftarMhs.add(e: "Dion");
11     daftarMhs.add(e: "Ending");
12     daftarMhs.add(e: "Fifin");
13     daftarMhs.add(e: "Gesit");
14     daftarMhs.add(e: "Heru");
15     daftarMhs.add(e: "Irma");
16     daftarMhs.add(e: "Janti");
17     System.out.println("N = " + daftarMhs.size());
18     for (int i=0; i<= daftarMhs.size()-1; i++)
19     {
20         System.out.println(i + " " + daftarMhs.get(i));
21     }
22     System.out.println(x: "");
23 }
Run | Debug
24 public static void main(String args[])
25 {
26     new p4();
27 }
28 }
```

PROBLEMS 31 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | AZURE | COMMENTS

```
● PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDe
6bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p4'
N = 10
0 Agungbp
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifin
6 Gesit
7 Heru
8 Irma
9 Janti

○ PS D:\AREAS\Semester3\Struktur data\12>
```

Praktik 5

```
import java.util.LinkedList;
public class p5 {
{
    LinkedList daftarMhs = new LinkedList ();
    p5()
    {
        daftarMhs.add("Agungbp");
        daftarMhs.add("Bambang");
        daftarMhs.add("Cucuk");
        daftarMhs.add("Dion");
        daftarMhs.add("Ending");
        daftarMhs.add("Fifin");
        daftarMhs.add("Gesit");
        daftarMhs.add("Heru");
        daftarMhs.add("Irma");
        daftarMhs.add("Janti");
        System.out.println("N = " + daftarMhs.size());
        for (int i=0; i<= daftarMhs.size()-1; i++)
        {
            System.out.println(i + " " + daftarMhs.get(i));
        }
        System.out.println("");
        daftarMhs.addFirst("Amir");
        daftarMhs.addLast("Zaenal");
        System.out.println("N = " + daftarMhs.size());
        for (int i=0; i<= daftarMhs.size()-1; i++)
        {
            System.out.println(i + " " + daftarMhs.get(i));
        }
        System.out.println("");
        System.out.println("Mhs awal = " + daftarMhs.getFirst());
        System.out.println("Mhs akhir = " + daftarMhs.getLast());
    }
    public static void main(String args[])
    {
        new p5();
    }
}
```

PROBLEMS 45 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE COMMENTS

```
● PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+Sh
6bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p5'
```

```
N = 10
0 Agungbp
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifin
6 Gesit
7 Heru
8 Irma
9 Janti
```

```
N = 12
0 Amir
1 Agungbp
2 Bambang
3 Cucuk
4 Dion
5 Ending
6 Fifin
7 Gesit
8 Heru
9 Irma
10 Janti
11 Zaenal
```

```
Mhs awal = Amir
Mhs akhir = Zaenal
```

```
○ PS D:\AREAS\Semester3\Struktur data\12>
```

Praktik 6

```
import java.util.*;
public class p6 {
    {
        LinkedList daftarMhs = new LinkedList ();
        p6() {
            t
            {
                daftarMhs.add("Agungbp");
                daftarMhs.add("Bambang");
                daftarMhs.add("Cucuk");
                daftarMhs.add("Dion");
                daftarMhs.add("Ending");
                daftarMhs.add("Fifin");
                daftarMhs.add("Gesit");
                daftarMhs.add("Heru");
                daftarMhs.add("Irma");
                daftarMhs.add("Janti");
                System.out.println("N = " + daftarMhs.size());
                for (int i=0; i<= daftarMhs.size()-1; i++)
                {
                    System.out.println(i + " " + daftarMhs.get(i));
                }
                System.out.println("");
                daftarMhs.remove(5);
                System.out.println("N = " + daftarMhs.size());
                for (int i=0; i<= daftarMhs.size()-1; i++)
                {
                    System.out.println(i + " " + daftarMhs.get(i));
                }
                System.out.println("");
                daftarMhs.removeFirst();
                daftarMhs.removeLast ();
                System.out.println("N = " + daftarMhs.size());
                for (int i=0; i<= daftarMhs.size()-1; i++)
                {
                    System.out.println(i + " " + daftarMhs.get(i));
                }
                System.out.println("");
            }
        }
        public static void main(String args[])
        {
            new p6();
        }
    }
}
```

PROBLEMS 57 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE COMMENTS

```
● PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe' -
6bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p6'
```

```
N = 10
0 Agungbp
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifin
6 Gesit
7 Heru
8 Irma
9 Janti
```

```
N = 9
0 Agungbp
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Gesit
6 Heru
7 Irma
8 Janti
```

```
N = 7
0 Bambang
1 Cucuk
2 Dion
3 Ending
4 Gesit
5 Heru
6 Irma
```

```
○ PS D:\AREAS\Semester3\Struktur data\12> |
```

Praktik 7

```
import java.util.LinkedList ;
public class p7 {
{
    LinkedList daftarMhs = new LinkedList ();
    p7() {
        {
            daftarMhs.add("Agungbp");
            daftarMhs.add("Bambang");
            daftarMhs.add("Cucuk");
            daftarMhs.add("Dion");
            daftarMhs.add("Ending");
            daftarMhs.add("Fifin");
            daftarMhs.add("Gesit");
            daftarMhs.add("Heru");
            daftarMhs.add("Irma");
            daftarMhs.add("Janti");
            System.out.println("");
            System.out.println("N = " + daftarMhs.size());
            for (int i=0; i<= daftarMhs.size()-1; i++)
            {
                System.out.println(i + " " + daftarMhs.get(i));
            }
            System.out.println("");
            daftarMhs.set(0,"Parmin");
            System.out.println("N = " + daftarMhs.size());
            for (int i=0; i<= daftarMhs.size()-1; i++)
            {
                System.out.println(i + " " + daftarMhs.get(i));
            }
            System.out.println("");
        }
    }
    public static void main(String args[])
    {
        new p7();
    }
}
```

```
PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe' -cp 'D:\AREAS\Semester3\Struktur data\12\6bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p7'
```

```
0 N = 10
0 Agungbp
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifi
6 Gesit
7 Heru
N = 10
0 Agungbp
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifi
6 Gesit
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifi
6 Gesit
7 Heru
8 Irma
9 Janti
4 Ending
5 Fifi
6 Gesit
7 Heru
8 Irma
9 Janti
7 Heru
8 Irma
9 Janti
9 Janti
```

```
N = 10
0 Parmin
1 Bambang
2 Cucuk
3 Dion
4 Ending
5 Fifi
6 Gesit
7 Heru
8 Irma
9 Janti
```

\

Praktik 8

Welcome

Snapcode

p8.java 9+

p7.java 9+

p4.java 9+

p8.java > p8 > p8()

```
1 import java.util.LinkedList;
2 public class p8{
3     LinkedList daftarMhs = new LinkedList();
4     p8()
5     {
6         daftarMhs.add(e: "Agungbp");
7         daftarMhs.add(e: "Bambang");
8         daftarMhs.add(e: "Cucuk");
9         daftarMhs.add(e: "Dion");
10        daftarMhs.add(e: "Ending");
11        daftarMhs.add(e: "Fifin");
12        daftarMhs.add(e: "Gesit");
13        daftarMhs.add(e: "Heru");
14        daftarMhs.add(e: "Irma");
15        daftarMhs.add(e: "Janti");
16        System.out.println("N = " + daftarMhs.size());
17        for (int i=0; i<= daftarMhs.size()-1; i++)
18        {
19            System.out.println(i + " " + daftarMhs.get(i));
20        }
21        System.out.println(x: "");
22        System.out.println(daftarMhs.contains(o: "Heru"));
23        System.out.println("Heru berada di posisi ke = " +
24        daftarMhs.indexOf(o: "Heru"));
25        System.out.println(x: "");
26    }
27    public static void main(String args[])
28    {
29        new p8(); } }
```

Run | Debug

PROBLEMS 82

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

AZURE

COMMENTS

PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe

N = 10

0 Agungbp

1 Bambang

2 Cucuk

2 Cucuk

2 Cucuk

3 Dion

4 Ending

5 Fifin

6 Gesit

7 Heru

8 Irma

9 Janti

true

Heru berada di posisi ke = 7

PS D:\AREAS\Semester3\Struktur data\12>

Praktik 9

```
import java.util.LinkedList ;
public class p9 {
    {
        s
        LinkedList daftarMh = new LinkedList ();
        p9()
        {
            s
            t
            {
                daftarMh .add("Agung");
                daftarMh .add("Saban");
                daftarMh .add("Gucu");
                daftarMh .add("Dion");
                daftarMh .add("Teding");
                daftarMh .add("Pifi");
                daftarMh .add("Deci");
                daftarMh .add("Haru");
                daftarMh .add("Trea");
                daftarMh .add("Jant");
                System.out.println ("M = " + daftarMh .size());
                for (int i=0; i<daftarMh .size()-1; i++)
                {
                    s
                    System.out.println (i + " " + daftarMh .get(i));
                }
                n
                System.out.println ("");
                daftarMh .clear ();
                System.out.println ("Clear selesai dijalankan... ");
                System.out.println ("M = " + daftarMh .size());
                for (int i=0; i<daftarMh .size()-1; i++)
                {
                    s
                    System.out.println (i + " " + daftarMh .get(i));
                }
                n
                System.out.println ("");
                if (daftarMh .isEmpty ()==true)
                { System.out.println ("linkedlist kosong ");
                }
                n
                g
            }
            else
            { System.out.println ("Linkedlist is ");
            }
            n
            t
        }
    }
    public static void main(String args[])
    {
        new p9();
    }
}
```

```
PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-90\redhat.java\jdt_ws\12_5638fb46\bin' 'p9'
```

```
N = 10
```

```
0 Agungbp
```

```
0 Agungbp
```

```
1 Bambang
```

```
1 Bambang
```

```
2 Cucuk
```

```
2 Cucuk
```

```
3 Dion
```

```
3 Dion
```

```
4 Ending
```

```
5 Fifi
```

```
6 Gesit
```

```
5 Fifi
```

```
6 Gesit
```

```
5 Fifi
```

```
5 Fifi
```

```
5 Fifi
```

```
6 Gesit
```

```
5 Fifi
```

```
5 Fifi
```

```
5 Fifi
```

```
5 Fifi
```

```
6 Gesit
```

```
7 Heru
```

```
8 Irma
```

```
9 Janti
```

```
Clear selesai dijalankan...
```

```
N = 0
```

```
Linkedlist kosong
```

```
PS D:\AREAS\Semester3\Struktur data\12>
```

LATIHAN

Latihan 1

```

import java.util.LinkedList ;
        t
public class p9 {

    LinkedList daftarMhs = new LinkedList ();
    t
    p9() {
        // Data Mahasiswa (Tipe Strin
        daftarMhs.add("Agungbp");
        daftarMhs.add("Bambang");
        daftarMhs.add("Cucuk");
        daftarMhs.add("Dion");
        daftarMhs.add("Ending");
        daftarMhs.add("Fifin");
        daftarMhs.add("Gesit");
        daftarMhs.add("Heru");
        daftarMhs.add("Irma");
        daftarMhs.add("Janti");

        daftarMhs.add(new Integer(10));
        daftarMhs.add(new Float(15.5f));
        daftarMhs.add(new Long(1000000L));

        System.out.println("N = " + daftarMhs.size());

        for (int i = 0; i <= daftarMhs.size() - 1; i++) {
            System.out.println(i + " " + daftarMhs.get(i));
        }

        System.out.println("");

        daftarMhs.clear();
        System.out.println("Clear selesai dijalankan... ");
        "
        System.out.println("N = " + daftarMhs.size());

        for (int i = 0; i <= daftarMhs.size() - 1; i++) {
            System.out.println(i + " " + daftarMhs.get(i));
        }

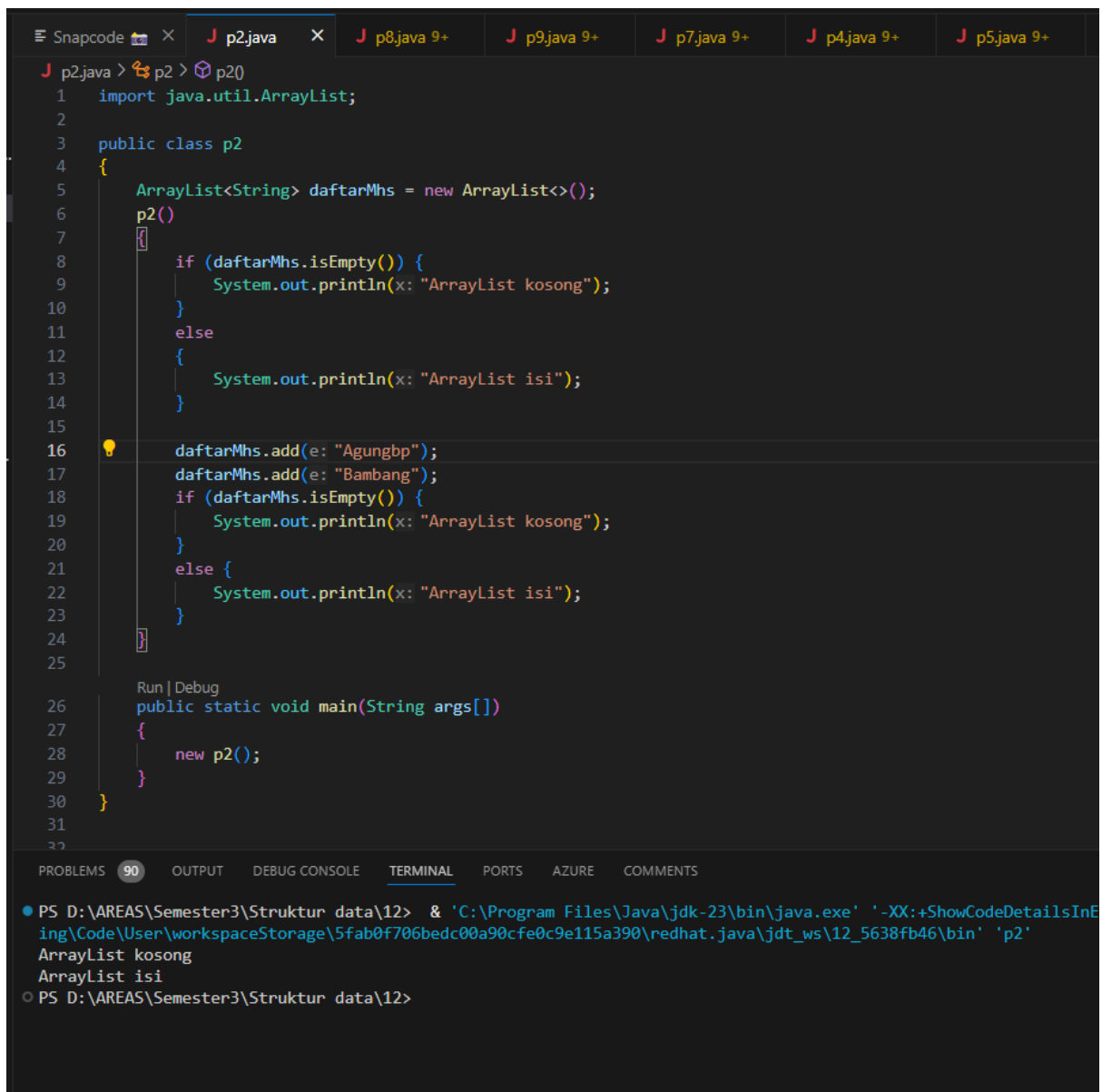
        System.out.println("");

        if (daftarMhs.isEmpty() == true) {
            System.out.println("Linkedlist kosong ");
        } else {
            g"
            System.out.println("Linkedlist is ");
        }
        i"
    }

    public static void main(String args[]) {

        new p9();
    }
}

```

The screenshot shows an IDE with a Java file named `p2.java`. The code defines a class `p2` with a constructor `p2()` and a `main` method. The constructor initializes an `ArrayList` named `daftarMhs` and checks if it is empty. The `main` method creates an instance of `p2`. The terminal output shows the execution results.

```
1 import java.util.ArrayList;
2
3 public class p2
4 {
5     ArrayList<String> daftarMhs = new ArrayList<>();
6     p2()
7     {
8         if (daftarMhs.isEmpty()) {
9             System.out.println(x: "ArrayList kosong");
10        }
11        else
12        {
13            System.out.println(x: "ArrayList isi");
14        }
15
16        daftarMhs.add(e: "Agungbp");
17        daftarMhs.add(e: "Bambang");
18        if (daftarMhs.isEmpty()) {
19            System.out.println(x: "ArrayList kosong");
20        }
21        else {
22            System.out.println(x: "ArrayList isi");
23        }
24    }
25
26    Run | Debug
27    public static void main(String args[])
28    {
29        new p2();
30    }
31 }
```

Terminal Output:

```
PS D:\AREAS\Semester3\Struktur data\12> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInE
ing\Code\User\workspaceStorage\5fab0f706bedc00a90cfe0c9e115a390\redhat.java\jdt_ws\12_5638fb46\bin' 'p2'
ArrayList kosong
ArrayList isi
PS D:\AREAS\Semester3\Struktur data\12>
```

TUGAS

Tugas 1

-